

Roll No.

TBT-504(c)

B. TECH. (BIOTECHNOLOGY)

(FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

GENOME EDITING TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What do you understand by homologous recombination ? Describe any *one* repair pathway that work during recombination. 10 Marks (CO1)

OR

- (b) What are DNA repair pathways ? Elaborate the pathway which activates during double strand break if homology is not available.

10 Marks (CO1)

2. (a) Describe Cre-loxP recombinase system of recombination.

10 Marks (CO1)

OR

- (b) Describe FLP-FRT recombinase system of recombination.

10 Marks (CO1)

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(2)

3. (a) What is RNA silencing ? How siRNA mediate gene silencing ?

10 Marks (CO1/CO2)

OR

- (b) What is RNA interference ? How miRNA facilitate RNA interference ?

10 Marks (CO1/CO2)

4. (a) What are Nucleases ? Describe different types of nucleases.

10 Marks (CO2)

OR

- (b) Explain the working of Zn finger nucleases and TALENS.

10 Marks (CO2)

5. (a) Describe CRISPR-Cas9 system of *S. pyogenes*.

10 Marks (CO2)

OR

- (b) What is CRISPR ? What are its components ? How do bacteria protect its own genome from Cas9 ?

10 Marks (CO2)